Probability - Homework 1

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1.2-17

(a)

To draw 4 cards with the same face value, there's 13 possibilities, then draw one from the rest of cards, there's $52-4=48$ possibilities (By definition, it's impossible to have same value with the previous 4 cards.), Hence, the probability is:

$$\frac{13\times 48}{\binom{52}{5}}=\frac{1}{4165}$$

(b)

We can choose two type of from 13 face values, and for each face value we have 4 colors to choose, then we can choose either 2 or 3 cards from them (but not for both types). (btw, we should time a extra 2, for instance, if we need to construct a full-house using 2 and 3, we can draw 2 cards from face value 2 and 3 cards from face value 3, or draw 3 cards from face value 2 and 2 cards from face value 3, there's two possibilities.) Thus, the probability is:

$$\frac{\binom{13}{2}\binom{4}{2}\binom{4}{3}\times 2}{\binom{52}{5}}=\frac{6}{4165}$$

(c)

Choose 3 ranks from 13 ($\binom{13}{3}$), then decide which of the 3 ranks forms the triple ($\times 3$, since any of the 3 chosen ranks can be assigned the triple role), choose 3 suits for it ($\binom{4}{3}$), and choose 1 suit for each of the remaining 2 kicker ranks ($\binom{4}{1}^2$). Thus, the probability is:

$$\frac{\binom{13}{3}\binom{4}{3}(\binom{4}{1})^2\times 3}{\binom{52}{5}}=\frac{88}{4165}$$

(d)

Choose 3 ranks from 13 ($\binom{13}{3}$), then decide which of the 3 ranks is the single kicker ($\times 3$, since any of the 3 chosen ranks can be assigned the kicker role), choose 1 suit for it ($\binom{4}{1}$), and choose 2 suits for each of the two pairs ($\binom{4}{2}^2$). Thus, the probability is:

$$\frac{\binom{13}{3}\binom{4}{1}(\binom{4}{2})^2\times 3}{\binom{52}{5}}=\frac{198}{4165}$$

(e)

Choose 4 ranks from 13 ($\binom{13}{4}$), then decide which of the 4 ranks forms the pair ($\times 4$, since any of the 4 chosen ranks can be assigned the pair role), choose 2 suits for it ($\binom{4}{2}$), and choose 1 suit for each of the 3 remaining kicker ranks ($\binom{4}{1}^3$). Thus, the probability is:

$$\frac{\binom{13}{4}\binom{4}{2}(\binom{4}{1})^3 \times 4}{\binom{52}{5}} = \frac{352}{833}$$

1.3-13

(a)

Let's denoted (a, b) as: The first dice rolls out a and the second dice rolls out b. There's 36 outcomes:
(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)
(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)
(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)
(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)
(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)
(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)

(b)

By (a), there are 6 possibilities of rolling out the sum of 7, and 2 possibilities for 11.

Hence, the probability for this question is $\frac{6+2}{36} = \frac{2}{9}$.

(c)

There are 5 possibilities of having sum 8, and 6 possibilities of having sum 7. Hence, we have:

$$P(8|7 \text{ or } 8) = \frac{5}{5+6} = \frac{5}{11}$$

(d)

There're 5 possibilities of having sum 8, $P(8) = \frac{5}{36}$.

By (b), $P(8|7 \text{ or } 8) = \frac{5}{11}$.

Hence, $P(8)P(8|7 \text{ or } 8) = \frac{5}{36} \times \frac{5}{11}$.

(e)

Similarily, we can consider the case of winning when rolling 4,5,6,8,9,10 and win.

$$\begin{aligned}
P(4)P(4|7\text{ or }4) &= \frac{3}{36}\times\frac{3}{9} = \frac{1}{36} \\
P(5)P(5|7\text{ or }5) &= \frac{4}{36}\times\frac{4}{10} = \frac{2}{45} \\
P(6)P(6|7\text{ or }6) &= \frac{5}{36}\times\frac{5}{11} = \frac{25}{396} \\
P(8)P(8|7\text{ or }8) &= \frac{5}{36}\times\frac{5}{11} = \frac{25}{396} \\
P(9)P(9|7\text{ or }9) &= \frac{4}{36}\times\frac{4}{10} = \frac{2}{45} \\
P(10)P(10|7\text{ or }10) &= \frac{3}{36}\times\frac{3}{9} = \frac{1}{36}
\end{aligned}$$

By adding the results above and the probability of winning at first roll, the answer is:

$$\frac{2}{9} + (\frac{1}{36} + \frac{2}{45} + \frac{25}{396} + \frac{25}{396} + \frac{2}{45} + \frac{1}{36}) \approx 0.49293$$

1.4-15

Define the random variable X as: The fourth white balls appearing at X-th round. And P be the probability function.

(a)

With replacement, the probability of taking white balls p is $\frac{1}{2}$. Also we can consider the experiment as a bernoulli trial. This is an problem of negative binomial distribution.

After a sequence of taking balls, we takes r-th white balls after taking k red balls, and the probability of taking white balls (in each round with replacement) is p, the probability is:

$$\binom{r+k-1}{k}(1-p)^k p^r$$

Hence, the answers can be obtained easily as below, there $p = \frac{1}{2}$ (Denote the probability of taking white ball):

$$\begin{aligned}
P(X=4) &= \binom{3}{0}(1-p)^0 p^4 = \frac{1}{16} \\
P(X=5) &= \binom{4}{1}(1-p)^1 p^4 = \frac{1}{8} \\
P(X=6) &= \binom{5}{2}(1-p)^2 p^4 = \frac{5}{32} \\
P(X=7) &= \binom{6}{3}(1-p)^3 p^4 = \frac{5}{32}
\end{aligned}$$

(b)

We can consider the results as the permutations of 10 white balls and 10 red balls. And if we take d-th white ball at r-th round (where $d \leq r$), the number of permutations is:

$$\frac{(r-1)!}{(d-1)!(r-d)!} \times \frac{(20-r)!}{(10-d)!(10-r+d)!} = \binom{r-1}{d-1} \times \binom{20-r}{10-d}$$

The total number of permutations of the 20 balls is :

$$\frac{20!}{10!10!} = \binom{20}{10}$$

By the formula above, we can easily obtain all the answers:

$$\begin{aligned}
P(X=4) &= \frac{1 \times 8008}{184756} = \frac{14}{323} \\
P(X=5) &= \frac{4 \times 5005}{184756} = \frac{35}{323} \\
P(X=6) &= \frac{10 \times 3003}{184756} = \frac{105}{646} \\
P(X=7) &= \frac{20 \times 1716}{184756} = \frac{60}{323}
\end{aligned}$$

(c)

Yes.

The World Series is best modeled as sampling **with replacement** because a team's probability of winning remains relatively constant; winning one game doesn't "deplete" their ability to win others. By setting $p = 0.5$ (assuming teams are evenly matched), the series length follows a Negative Binomial distribution. We multiply the single-team probability by 2 to account for symmetry (It's 1), which means either the American or National League can reach four wins first.

1.5-4

Consider 4 partitions (age 16-25, 26-50, 51-65, 66-90, respectively) as B_1, B_2, B_3, B_4, and A be the event of accident. By Bayes' Theorem we obtain the answer:

$$\begin{aligned}
P(B_1|A) &= \frac{P(A|B_1)P(B_1)}{P(A|B_1)P(B_1) + P(A|B_2)P(B_2) + P(A|B_3)P(B_3) + P(A|B_4)P(B_4)} \\
&= \frac{0.05 \times 0.10}{0.05 \times 0.10 + 0.02 \times 0.55 + 0.03 \times 0.20 + 0.04 \times 0.15} \\
&= 0.178571428571 \approx 0.1786
\end{aligned}$$